1

Kähler Manifolds

Some facts to eventually include:

1. In real manifolds, except in dimension 4 manifolds have a unique real structure. Usually there are infinitely many (often uncountably infinitely many) complex structures on a topological manifold. Think of elliptic curves. This inspires Teichmüller theory and creates interesting moduli spaces and is the source of some of the easiest example of stacks

1.1 Basics of Complex Manifolds

Let $U \subseteq \mathbb{C}^n$. Then $f : U \to \mathbb{C}$ be holomorphic if it is holomorphic in each variable, and $F = (f_1, ..., f_n) : U \to \mathbb{C}^n$ is holomorphic if it is also holomorphic in each component. By identifying $\mathbb{C}^n \cong \mathbb{R}^{2n}$, let $z_j = x_j + iy_j$ and $f_k = u_k = iv_k$ for $1 \leq j, k \leq n$. Certainly u_k, v_k are smooth in $x_1, y_1, ...x_n, y_n$ and satisfy the Cauchy Riemann equations:

$$\frac{\partial u_k}{\partial x_j} = \frac{\partial v_k}{\partial y_j} \qquad \text{and} \qquad \frac{\partial u_k}{\partial y_j} = -\frac{\partial v_k}{\partial x_j}$$

Conversely, given a smooth map $(u_1, v_1, ..., u_n, v_n) : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ where each pair u_i, v_i satisfies the Cauchy-Riemann equations, then the map $(u_1 + iv_1, ..., u_n + iv_n)$ is holomorphic. In terms of DF, given that the tangent space $T_p(\mathbb{R}^{2n})$ is written in coordinates:

$$\left\{ \frac{\partial}{\partial x_1}, ..., \frac{\partial}{\partial x_n}, \frac{\partial}{\partial y_1}, ..., \frac{\partial}{\partial y_n} \right\}$$

and the $T_{F(p)}(\mathbb{R}^{2n})$ is written in basis:

$$\left\{ \frac{\partial}{\partial u_1}, ..., \frac{\partial}{\partial u_n}, \frac{\partial}{\partial v_1}, ..., \frac{\partial}{\partial v_n} \right\}$$

Then we get the familiar:

$$DF(p) = \begin{pmatrix} A & -B \\ B & A \end{pmatrix} \tag{1.1}$$

where A, B are the matrices for the $\frac{\partial}{\partial x_j}$ and $\frac{\partial}{\partial y_j}$ respectively; let us call this matrix the *CR matrix* (for Cauchy-Riemann, as it represents the relationship). This can be verified computationally (without needing to find a nice basis) by checking whether the such a matrix commutes with:

$$J := \begin{pmatrix} 0 & -I_n \\ I_n & 0 \end{pmatrix}$$

Now upgrading this to a manifold:

Definition 1.1.1: Complex Manifold

A topological manifold M is a *complex manifold* if there exists a *complex atlas* or *complex structure* (U_a, φ_α) where:

1. $M = \bigcup_\alpha U_\alpha$ is an open covering
2. $\varphi_\alpha : U_\alpha \to V_\alpha \subseteq \mathbb{C}^n$ is a homeomorphism (where by [2] n is independent of α)
3. If $U_\alpha \cap U - \beta \neq \emptyset$ then

   $$\varphi_\beta \cap \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$$

   is a biholomorphism.

If M, N are complex manifolds, $f : M \to N$ is a holomorphic map if it is holomorphic in coordinates. The map f is biholomorphic if it has a holomorphic inverse.

Given open $U \subseteq M$, let $\mathcal{O}(U) = \{f : U \to \mathbb{C} \mid f \text{ holomorphic}\}$ and similarly $\mathcal{O}^*(U)$ be the set of nowhere-vanishing (nowhere-zero) holomorphic functions.

complex manifolds are canonically oriented, not just orientable.

Example 1.1.2: Complex Manifolds

The details that any of these are complex manifolds follows almost entirely from differential geometry, see [2].

1. $\mathbb{C}^n$ and every open subset naturally is a complex manifold
2. A special open subset is $\mathrm{GL}(n, \mathbb{C})$.
3. Complex projective space $\mathcal{P}^n$ is a compact complex manifold
4. the complex Grassmannian $G(k, n)$ of k-dimensional subspaces of $\mathbb{C}^n$ are compact complex manifolds. Note that $G(k, n)$ can be viewed as a quotient of $V(k, n = \{W \in M_{n\times k}(\mathbb{C}) \mid \mathrm{rank}(W) = k\}$ by:

   $$G(k, n) := \frac{V(k, n)}{\mathrm{GL}(k, \mathbb{C})}$$

   where the multiplication is on the right.

5. Complex Lie groups
6. (complex) varieties. Elliptic curves being a source of many examples.
7.

Proposition 1.1.3: Maximum Principle On Complex Manifolds

let M be a compact, connected, complex manifold and $f : M \to \mathbb{C}$ holomorphic. Then f is constant

***Proof*:**
Use the fact that the maximum modulus principle holds for holomorphic functions of several variables and analytic continuation (see [3])

Let $F : M \to N$ be a holomorphic map and choose coordinates so that $\tilde{F} : U \to V$ where $U \subseteq \mathbb{C}^n$, $V \subseteq \mathbb{C}^n$; let $\tilde{F} = (f_1, ..., f_m)$ in coordinates. Then:

$$D\tilde{F}(p) : U \to V \qquad D\tilde{F}(p)(v) = \left(\frac{\partial f_i}{\partial z_j}(p)\right) \cdot v$$

By the Cauchy-Riemann equations, the matrix representation is of the form in equation (1.1). It is not hard to see that the inverse respects this form, recovering the complex inverse function theorem for manifolds. Naturally, this also recovers the implicit version, and more generally the Rank theorem, see [2]. Given we have the rank theorem, we can recover similar results for submanifolds. Let $F : M \to N$ be a holomorphic map. Then:

$$\mathrm{rank}_p(F) = \mathrm{rank}(D(\psi \circ F \circ \varphi^{-1})(\varphi(p))$$

for a choice of local coordinates[1]. Then:

Theorem 1.1.4: Complex Submanifold

Let $F : M \to N$ be a holomorphic map, $q \in F(M)$, $X = F^{-1}(\{q\})$. Suppose $\mathrm{rank}_x(F) = k$ for all $x \in U$ where U contains X. Then X is a complex manifold and:

$$\mathrm{codim} X = k$$

***Proof*:**
exactly the same as in [2]

There is a *pivotal* difference between the existence of real and complex submanifolds

Proposition 1.1.5: All compact complex manifolds of $\mathbb{C}^n$ are trivial

There are no compact complex submanifold of $\mathbb{C}^n$ of dimension greater than 0

[1] the rank is independent of this choice

***Proof*:**
Let $M \subseteq \mathbb{C}^n$ be a submanifold. Then each coordinate function z_i restricts o a holomorphic function on M. But M is compact, hence by proposition 1.1.3 z_i must be locally constant, hence $\dim M = 0$.

Thus, there cannot be a Whitney Embedding theorem for holomorphic manifolds. We shall prove in ref:HERE that there is the *Kodaira Embedding Theorem* which gives the conditions for a compact complex manifold to be embedded in $\mathbb{P}^n$.

To define more complicated structure on a complex manifold, we need to have the appropriate complex vector bundle theory:

Definition 1.1.6: Complex Vector Bundle

A *holomorphic vector bundle* E over a complex manifold M is a complex manifold E together with a holomorphic map $\pi : E \to M$ such that:

1. For each $x \in M$, the fiber $E_x = \pi^{-1}(x)$ is a complex vector space of dimension d (the rank of E).
2. There exists an open covering $\{U_\alpha\}$ of M and biholomorphic maps

$$\Phi_\alpha : \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}^d$$

   such that

   (a) $p_1(\Phi_\alpha(x)) = x$ for all $x \in U$, where $p_1 : U_\alpha \times \mathbb{C}^d \to U_\alpha$ denotes projection on the first factor, and

   (b) For every $x \in U_\alpha$ the map $p_2 \circ \Phi_\alpha|_{E_x} : E_x \to \mathbb{C}^d$ is an isomorphism of complex vector spaces.

Call E the total space of the bundle and M its base. The covering $\{U_\alpha\}$ is called a local trivializing cover of M and the biholomorphisms $\{\Phi_\alpha\}$ local trivializations. When $d = 1$ we often refer to E as a line bundle.

If $U \subseteq M$, then a holomorphic map $\sigma : U \to E$ such that $\pi \circ \sigma = \mathrm{id}|_U$ is called a *holomorphic section* of M. The collection of all section of E of an open set U is denoted $\Gamma(U, E)$ and these are each a $\mathcal{O}(U)$-modules.

A holomorphic vector bundle can also be described using transition functions, namely, given a covering of M by open sets U_α take holomorphic maps:

$$g_{\alpha\beta} : U_\alpha \cap U_\beta \to \mathrm{GL}(d, \mathbb{C})$$

such that

$$g_{\alpha\beta} \cdot g_{\beta\gamma} = g_{\alpha\gamma}$$

on $U_\alpha \cap U_\beta \cap U_\gamma$. The maps $g_{\alpha\beta}$ are defined by the commutative diagram:

$$\begin{tikzcd} & \pi^{-1}(U_\alpha \cap U_\beta) \arrow[dl, "\Phi_\beta"'] \arrow[dr, "\Phi_\alpha"] & \\ (U_\alpha \cap U_\beta) \times \mathbb{C}^d \arrow[rr, "(\mathrm{id}, g_{\alpha\beta})"] & & (U_\alpha \cap U_\beta) \times \mathbb{C}^d \end{tikzcd}$$

A holomorphic line bundle over M is thus given by a collection $\{U_\alpha, g_{\alpha\beta}\}$, where U_α is an open cover of M and the $\{g_{\alpha\beta}\}$ are *nowhere-zero* holomorphic functions defined on $U_\alpha \cap U_\beta$, i.e. $g_{\alpha\beta} \in \mathcal{O}^*(U_\alpha \cap U_\beta)$ satisfying the cocycle condition.

If $F : N \to M$ is a holomorphic map and $\pi : E \to M$ a holomorphic vector bundle, then we can define a pullback:

$$h_{\alpha\beta} : F^{-1}(U_\alpha) \cap F^{-1}(U_\beta) \to \mathrm{GL}(d, \mathbb{C}) \qquad h_{\alpha\beta} = g_{\alpha\beta} \circ F$$

for local trivialization given by U_α's and transition functions $g_{\alpha\beta}$. This gives a diagram:

$$\begin{tikzcd} F^*(E) \arrow[r, "\overline{F}"] \arrow[d, "\pi^*"'] & E \arrow[d, "\pi"] \\ N \arrow[r, "F"] & M \end{tikzcd}$$

If we look at line bundles in particular, as their tensor product preserves dimension (since $1 \cdot 1 = 1$), they are of particular interest. If L, L' are two line bundle cover X subject to the same local trivialization with transition functions $g_{\alpha\beta}, g'_{\alpha\beta}$, then:

$$h_{\alpha\beta} = g_{\alpha\beta} g'_{\alpha\beta}$$

define a line bundle, and is denoted $L \otimes L'$. Similarly, the tensor product can be defined in the usual way and the transition function would be the above multiplication. The dual bundle of L, denoted L^{-1}, L^* or $\hom(L, \mathbb{C})$ is defined as

$$h_{\alpha\beta} = (g_{\alpha\beta})^{-1}$$

Then $L \otimes L^{-1}$ is the trivial bundle.

Example 1.1.7: Complex Vector Bundle

For now I'm relegating all examples to [1]. An important one to point from algebraic geometry is that the tautological bundle on $\mathbb{P}^n$ is denoted $\mathcal{O}(-1)$, and its dual $\mathcal{O}(1)$. By some work in algebraic geometry, it is easy to verify that this notation is justified as it corresponds to a hyperplane $H \cong \mathbb{P}^{n-1} \subseteq \mathbb{P}^n$. Note this hyperplane can be given as the zero-locus of a section of a line bundle.

A similar result can be done for $G(n, k)$.

Next, let $L \to M$ be a line bundle with local trivialization $\{U_\alpha, \varphi_\alpha\}_\alpha$. Then a section $\sigma : M \to L$ may be given by taking a collection of holomorphic function $f_\alpha \in \mathcal{O}(U_\alpha)$ and setting:

$$\sigma(x) = f_\alpha(x) \varphi_\alpha^{-1}(x, 1)$$

Thus, for any $x \in U_\alpha \cap U_\beta$:

$$f_\alpha(x) = g_{\alpha\beta}(x) \cdot f_\beta(x)$$

showing that line bundles are functions "up to" a transition factor on each intersection. We shall see in ref:HERE that all line bundles on $\mathbb{P}^n$ will be of the form $\mathcal{O}(d)$ that is homogeneous degree n polynomials.

Next, let us consider differential forms on complex manifolds. A natural place to start is by seeing how